THE LOCUS OF GEOMETRICALLY REDUCED FIBRES OF A FLAT PROPER FINITELY PRESENTED MORPHISM IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7016-7054, immediately following the statement at 7004-7014.

Editorial note. Eight cross-chapter pointers (Algebra, Limits of Spaces, Spaces over Fields, Topology) plus three same-chapter lemmas are recorded in assembly.external_references.

Editorial note. Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is not counted beside it.

Allowed standard prerequisites (editorial).

- Base change preservation of geometrically reduced fibres. chapter lemma lemma-base-change-fibres-geometrically-reduced, cited at proof line 7017. Makes the property checkable after base change, which is the first reduction.
- Constructibility of the geometrically reduced locus. chapter lemma lemma-geometrically-reduced-constructible, cited in the proof. Turns openness into a constructibility plus generisation problem.
- Proper flat over a discrete valuation ring with reduced special fibre. chapter lemma lemma-proper-flat-over-dvr-reduced-fibre, cited at proof line 7053. Supplies the local model case to which the argument reduces.
- Reducedness of a fibre after a finite field extension of the base. Spaces over Fields, lemmas spaces-over-fields-lemma-perfect-reduced and spaces-over-fields-lemma-geometrically-reduced-positive-characteristic, cited at proof lines 7046-7049. Produces the finite extension L of the fraction field over which the fibre stops being reduced.
- Dominated discrete valuation ring inside a finite extension. Algebra, lemma algebra-lemma-integral-closure-Dedekind, cited at proof lines 7050-7051. Finds a discrete valuation ring R' inside L dominating R, completing the reduction.
- Characterisation of closed sets in a Noetherian space. Topology, lemma topology-lemma-characterize-closed-Noetherian, cited in the proof. Turns generisation stability into openness in the Noetherian situation.

Source reference. Original reference 09IG: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E09: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6916–6916. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0E0A: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6939–6939. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E0B: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6971–6971. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 054F: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SK: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08K0: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08K1: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0E03: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0E02: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0542: focused-seeds/geometry/S9-spaces-geom-reduced-open/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is flat, proper, and of finite presentation, then the set*

$$E = \{y \in |Y| : \textit{the fibre of } f : X \to Y \textit{ at } y \textit{ is geometrically reduced}\}$$

is open in $|Y|$.

Proof. By Lemma 0E09 formation of E commutes with base change. To check a subset of $|Y|$ is open, we may replace Y by the members of an étale covering. Thus we may assume Y is affine. Then Y is a cofiltered limit of affine schemes of finite type over $\mathbf{Z}$. Hence we can assume $X \to Y$ is the base change of $X_0 \to Y_0$ where Y_0 is the spectrum of a finite type $\mathbf{Z}$-algebra and $X_0 \to Y_0$ is flat and proper. See Limits of Spaces, Lemma 07SK, 08K0, and 08K1. Since the formation of E commutes with base change (see above), we may assume the base is Noetherian.

Assume Y is Noetherian. The set is constructible by Lemma 0E0A. Hence it suffices to show the set is stable under generalization (Topology, Lemma 0542). By Properties, Lemma 054F we reduce to the case where $Y = \mathrm{Spec}(R)$, R is a discrete valuation ring, and the closed fibre X_y is geometrically reduced. To show: the generic fibre X_η is geometrically reduced.

If not then there exists a finite extension L of the fraction field of R such that X_L is not reduced, see Spaces over Fields, Lemmas 0E02 (characteristic zero) and 0E03 (positive characteristic). There exists a discrete valuation ring $R' \subset L$ with fraction field L dominating R, see Algebra, Lemma 09IG. After replacing R by R' we reduce to Lemma 0E0B. □